

COMPUTER NETWORKS

Date: _____

NAME: ARIZ ALI

Roll Number: 24K-1044

ASSIGNMENT 01

SECTION: B (NUMERICALS)

Q1.

Root DNS RTT = 60ms

TLD DNS RTT = 50ms

Authoritative DNS RTT = 40ms

a)

$$\begin{aligned}\text{Total DNS Resolution Time} &= \text{RTT}_{\text{root}} + \text{RTT}_{\text{TLD}} + \text{RTT}_{\text{Auth}} \\ &= 60\text{ms} + 50\text{ms} + 40\text{ms} \\ &= 150\text{ms}.\end{aligned}$$

b) New Resolution Time = $\text{RTT}_{\text{Auth}} = 40\text{ms}$

$$\begin{aligned}\text{Time Saved} &= \text{Original} - \text{New} \\ &= 150\text{ms} - 40\text{ms} \\ &= 110\text{ms}\end{aligned}$$

Q2. Initial Cache Time = 10:00:00 AM

TTL = 300sec $\Rightarrow$ 5min

Cache Entry Expiration time = 10:05:00 AM

Uncached Resolution Time = 150ms

Cache Lookup time : 2ms

a) Yes, Cache Entry is still valid at 10:03:00 AM

$$\begin{aligned}\text{Time Saved} &= \text{Uncache Resolution Time} - \text{Cache lookup time} \\ &= 150\text{ms} - 2\text{ms} \\ &= 148\text{ms}\end{aligned}$$

b) No, Cache Entry is invalid at 10:07:00 AM,

TTL of 300second causes Entry to Expire at 10:05:00 AM. Since 10:07:00 AM is 7min (420 sec) after initial cache time, the Entry has Expired requires full Resolution Time.

Q3. File Size (L) = 12 MB $\Rightarrow$ 96 Mbits.

Link 1 Bandwidth (R_1) = 20 Mbps

Link 2 Bandwidth (R_2) = 100 Mbps

i- Transmission delay on Link 1

$$d_{t1} = \frac{L}{R_1} \Rightarrow \frac{96}{20} \Rightarrow 4.8 \text{ sec}$$

ii- Transmission delay on Link 2

$$d_{t2} = \frac{L}{R_2} = \frac{96}{100} = 0.96 \text{ sec}$$

$$\text{Total Transmission Delay} = d_{t1} + d_{t2} = 4.8 + 0.96 \Rightarrow 5.76 \text{ sec}$$

Q4. Packet Size (L) = 8000 bits

Link 1 Bandwidth = 40 Mbps = 40×10^6 bps

propagation delay = 20 ms $\Rightarrow$ 0.02 s

Link 2 : Bandwidth = 80 Mbps = 80×10^6 bps

propagation delay = 30 ms $\Rightarrow$ 0.03 sec

Link 3 : Bandwidth = 500 Mbps $\Rightarrow$ 500×10^6 bps

Propagation delay = 5 ms $\Rightarrow$ 0.005 s

$$d_{t1} = \frac{8000}{40 \times 10^6} \Rightarrow 0.0002 \text{ s} \Rightarrow 0.2 \text{ ms}$$

$$d_{t2} = \frac{8000}{80 \times 10^6} \Rightarrow 0.0001 \text{ s} \Rightarrow 0.1 \text{ ms}$$

$$d_{t3} = \frac{8000}{500 \times 10^6} \Rightarrow 0.000016 \text{ s} \Rightarrow 0.016 \text{ ms}$$

$$\text{Total delay}_{\text{tran}} \Rightarrow 0.2 + 0.1 + 0.016 \Rightarrow 0.316 \text{ ms}$$

Date: _____

$$\text{Total propagation delay} = 20 + 30 + 5 \Rightarrow 55 \text{ ms}$$

$$\begin{aligned}\text{End-to-End delay} &= \text{Transmission delay} + \text{Propagation delay} \\ &= 0.316 + 55 \\ &= 55.316 \text{ ms}\end{aligned}$$

Q5. Request Per hour (N) = 5000 req/hour
Average Object Size (S) = 400 kb
Cache hit Rate (h) = 75% $\Rightarrow$ 0.75

$$\begin{aligned}\text{a) Total Traffic} &= N \times S \Rightarrow 5000 \times 400 \\ &= 2000000 \text{ kb} \\ &= 2000 \text{ Mb} \Rightarrow 2 \text{ Gb}\end{aligned}$$

b) Since 75% of Request are served by Cache
~~Cache~~ Cache miss = 0.25

$$\begin{aligned}\text{Traffic to Original Server} &= \text{Total Traffic} \times (\text{Cache Miss}) \\ &= 2000 \text{ Mb} \times 0.25 \\ &\Rightarrow 500 \text{ Mb}\end{aligned}$$

$$\begin{aligned}\text{c) Traffic Saved} &= \text{Total Traffic} \times h \\ &= 2000 \text{ Mb} \times 0.75 \\ &= 1500 \text{ Mb} \Rightarrow 1.5 \text{ Gb}\end{aligned}$$